

Spanish Scientist Uses Bacteria to Repair 18th-century Artworks

Spanish scientist Pilar Bosch has found a way to use bacteria to repair 18th-century paintings.

Bosch, a microbiologist, tells Reuters she got the idea for the method while considering possibilities for a paper toward her doctoral degree. During this process, she discovered someone else's paper that suggested bacteria can be used to restore artworks.

At around the same time, Bosch learned that her mother was struggling to find a good method for restoring 18th-century paintings. The paintings, by Spanish artist Antonio Palomino, sat in Santos Juanes – one of the oldest churches in the city of Valencia.

Pilar Roig Picazo, 75, and her daughter Pilar Bosch Roig, 42, work together on the restoration process at Santos Juanes in Valencia, Spain, September 17, 2024. (REUTERS/Eva Manez)

Bosch said her mother, Pilar Roig, noted how difficult the restoration process at the church was. One reason was the glue used to remove the paintings from the walls in the 1960s. The substance was nearly impossible to remove completely.

"My mother had a very difficult problem to solve and I found a paper about bacteria used to clean frescoes in Italy," Bosch said. A fresco is a kind of painting that often appears across a large wall.

Bosch added she decided to write her doctoral paper on the same subject. That was more than ten years ago. Today, she gets to work with her mother on the church restoration. They are using the bacteria method to remove glue from artworks. The \$4.46 million project is being financed by local organizations.

Pilar Roig Picazo, 75, and her daughter Pilar Bosch Roig, 42, watch as restorers work on a mural painting removed from the Santos Juanes church in Valencia, Spain, September 17, 2024. (REUTERS/Eva Manez)

The process involves feeding small amounts of the glue – made from animal collagen – to the bacteria. After a time, the bacteria begins to naturally produce enzymes to break down the glue.

The mother-daughter team then mixes the bacteria with a natural algae-based substance, called a gel, and places it onto the paintings. They say that after about three hours, the gel mixture is removed, leaving the paintings glue-free.

Bosch's mother, now 75, noted that in the past, restorers of artworks had to use traditional cleaning methods that "took hours and damaged the painting." She noted that several other relatives in the family also work in art restoration. Bosch agreed, saying the work "certainly runs in the family."

Bosch has also used her bacteria removal process in other restoration projects in Italy and Spain. And, she is now training other scientists to use a different kind of bacteria to clear walls of painted graffiti.

I'm Bryan Lynn.

Reuters reported this story. Bryan Lynn adapted the report for VOA Learning English.

Vocabulary

restore - v. ซ่อมแซมสิ่งที่เก่า

church - n. โบสถ์, ศาสนสถานสำหรับนมัสการพระเจ้า

glue - n. กาว, สารที่ใช้ยึดสิ่งต่างๆ เข้าด้วยกัน

graffiti - n. กราฟฟิตี, ภาพวาดหรือข้อความที่เขียนบนกำแพงหรือที่สาธารณะ มักทำอย่างผิดกฎหมาย

Reading Comprehension Quiz

1. What specific type of artwork did Spanish scientist Pilar Bosch find a way to repair using bacteria?

- Ancient sculptures
- 18th-century paintings
- Medieval tapestries
- Modern digital art

2. Pilar Bosch's idea for using bacteria in art restoration initially came about during what academic pursuit?

- A research project for a master's degree
- Her studies in microbiology
- Considering possibilities for a paper toward her doctoral degree
- A collaborative project with other scientists

3. What specific type of scientific document gave Pilar Bosch the initial suggestion that bacteria could be used for artwork restoration?

- A textbook on microbiology

Her mother's restoration notes

Someone else's published paper

An ancient manuscript

4. The 18th-century paintings that Bosch's mother was struggling to restore were located in which specific place?

A national museum in Madrid

Santos Juanes church in Valencia

An art gallery in Italy

Her family's private collection

5. According to Bosch's mother, what was a significant factor making the restoration process at the church difficult?

The delicate nature of the paint

Lack of proper tools

The glue used to remove the paintings in the 1960s

Insufficient funding for the project

6. What is a 'fresco' as defined in the article?

A type of glue used in restoration

A large wall painting

A specific species of bacteria

A historical document

7. How long ago did Pilar Bosch decide to write her doctoral paper on the subject of using bacteria for restoration?

About five years ago

More than ten years ago

Recently, in the last year

During her undergraduate studies

8. The current \$4.46 million restoration project involving the mother-daughter team is being financed by which entities?

The Spanish government

International art foundations

Local organizations

Private family funds

9. What is the primary ingredient of the glue that is fed to the bacteria in the restoration process?

Synthetic polymers

Plant-based cellulose

Animal collagen

Mineral compounds

10. What do the bacteria naturally produce after being fed the glue to break it down?

A corrosive acid

Enzymes

A neutralizing agent

Bubbles to lift the glue

11. What substance is mixed with the bacteria before being applied to the paintings?

A synthetic solvent

A natural algae-based gel

Distilled water

A mild soap solution

12. Approximately how long is the gel mixture left on the paintings before it is removed?

About 30 minutes

About one hour

About three hours

Overnight

13. According to Bosch's mother, what was a negative consequence of traditional art cleaning methods?

They were too expensive

They required highly specialized equipment

They took hours and damaged the painting

They failed to remove the glue completely

14. What did Pilar Bosch state about art restoration within her family?

It is a new passion for everyone

It is a hobby for some members

The work 'certainly runs in the family'

Only she and her mother are involved

15. Besides her current project, what new application of bacteria-based cleaning is Pilar Bosch training other scientists for?

Cleaning historical documents

Restoring ancient textiles

Clearing walls of painted graffiti

Preserving archaeological artifacts